

Programming assignment

2025 Fall - 機器學習

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(1) 資料轉換

請將原始資料轉換為兩個監督式學習資料集：

(a) 分類 (Classification) 資料集

- 格式：(經度, 緯度, label)
- 規則：
 - 若溫度觀測值為 無效值 -999，則 label = 0。
 - 若溫度觀測值為 有效值，則 label = 1。

(b) 回歸 (Regression) 資料集

- 格式：(經度, 緯度, Value)
- 規則：
 - 僅保留有效的溫度觀測值（剔除所有 -999）。
 - value 為對應的攝氏溫度。

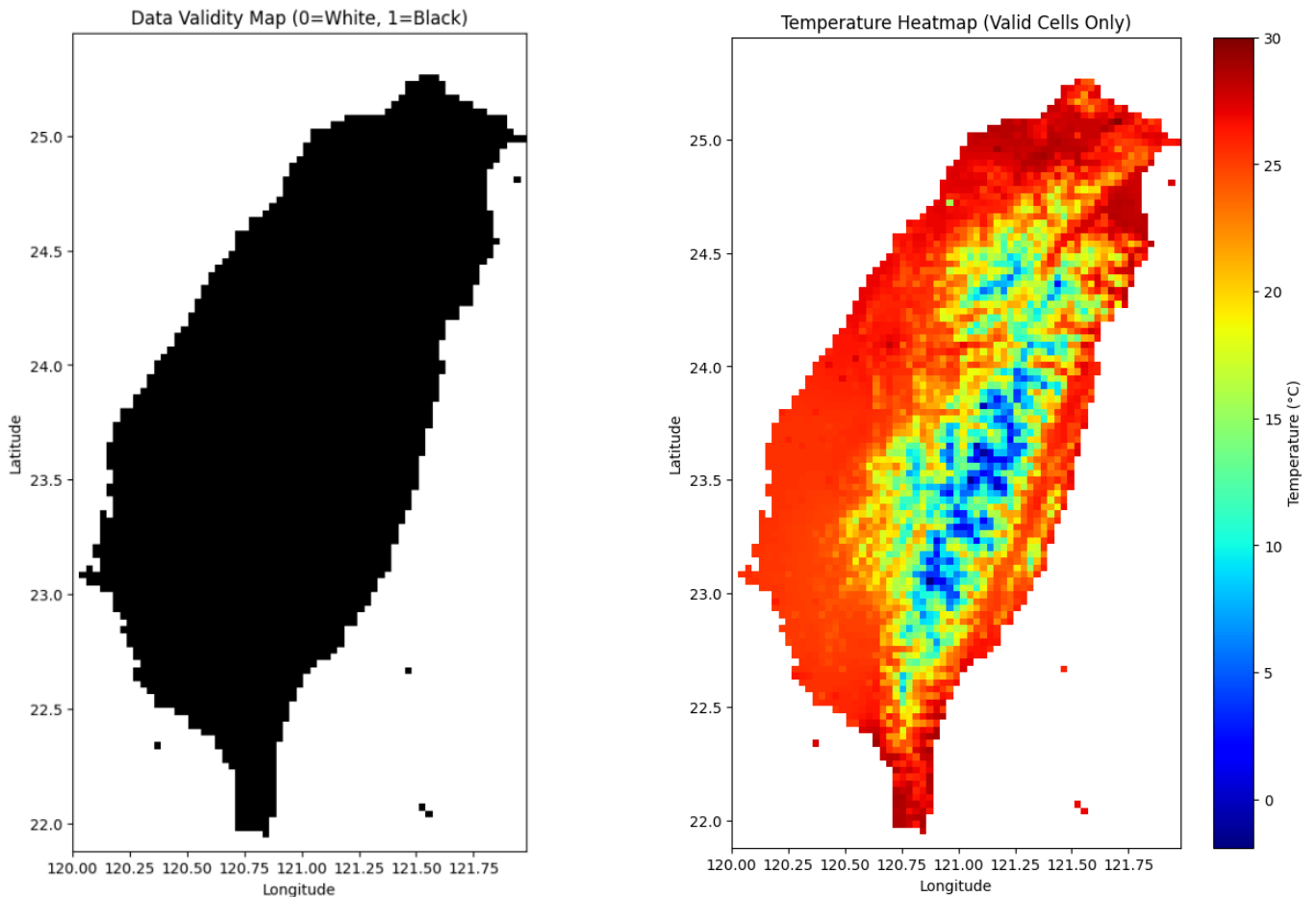
We use Python to read and clean weather data from an XML file, organizing the numbers into a grid with real locations. Then, we make two datasets: one for marking valid or invalid temperatures, and one for keeping only the real temperature values. Both datasets are saved as CSV files for machine learning use.

分類資料集前五筆

	Longitude	Latitude	Label
0	120.00	21.88	0
1	120.03	21.88	0
2	120.06	21.88	0
3	120.09	21.88	0
4	120.12	21.88	0

回歸資料集前五筆

	Longitude	Latitude	Value
0	120.84	21.94	28.1
1	120.72	21.97	28.6
2	120.75	21.97	28.6
3	120.78	21.97	27.8
4	120.81	21.97	26.5



These two images show the data validity and temperature distribution of meteorological observations. The left image uses black and white to represent data validity, where black indicates valid temperature measurements and white indicates invalid data. The right image displays a color heatmap illustrating temperature variations across valid grid points, with warmer colors indicating higher temperatures.

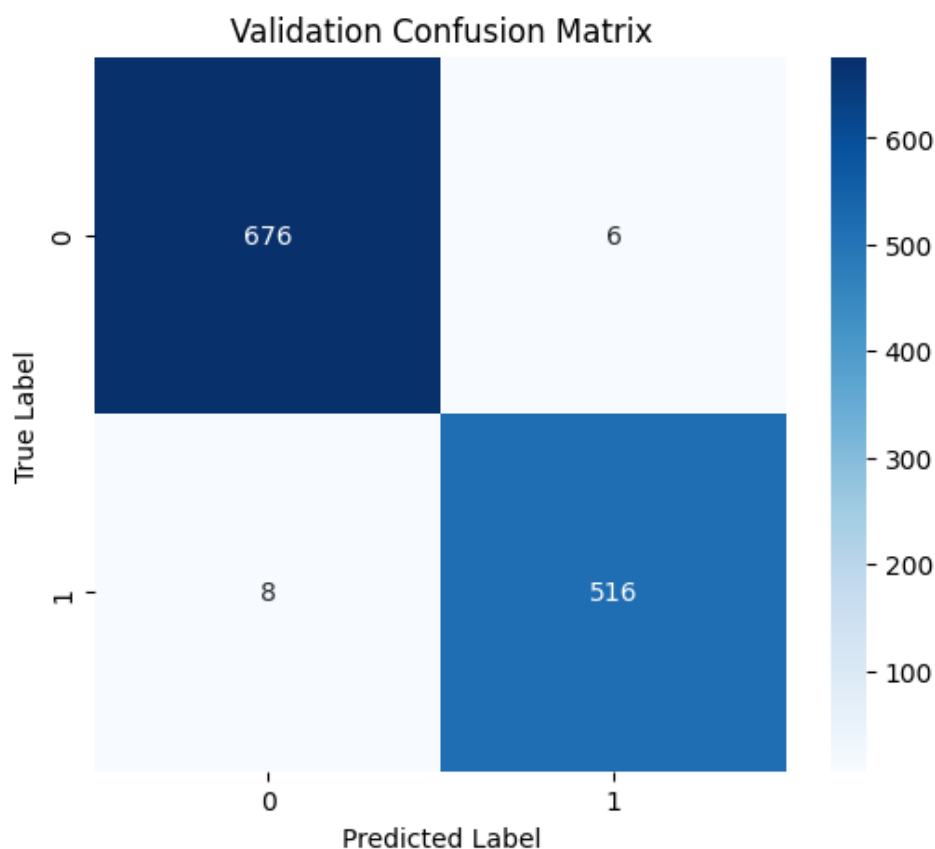
(2) 模型訓練

使用 (1) 中整理出的兩個資料集，分別訓練一個簡單的機器學習模型：

- **分類模型 (classification model) :**
以 (經度, 緯度) 預測格點資料是否為有效值 (0 或 1)。
- **回歸模型 (regression model) :**
以 (經度, 緯度) 預測對應的溫度觀測值。

The classification model code uses a Random Forest classifier with feature standardization, taking longitude and latitude as features to predict whether each grid point's data is valid (label 0 or 1). The data is split into 70% training, 15% validation, and 15% test sets. The model learns on the training set, tunes and evaluates on the validation set, and finally assesses performance on the test set.

Model evaluation uses a confusion matrix heatmap, showing the comparison between predicted and true labels, with color intensity reflecting the frequency of correct and incorrect classifications.

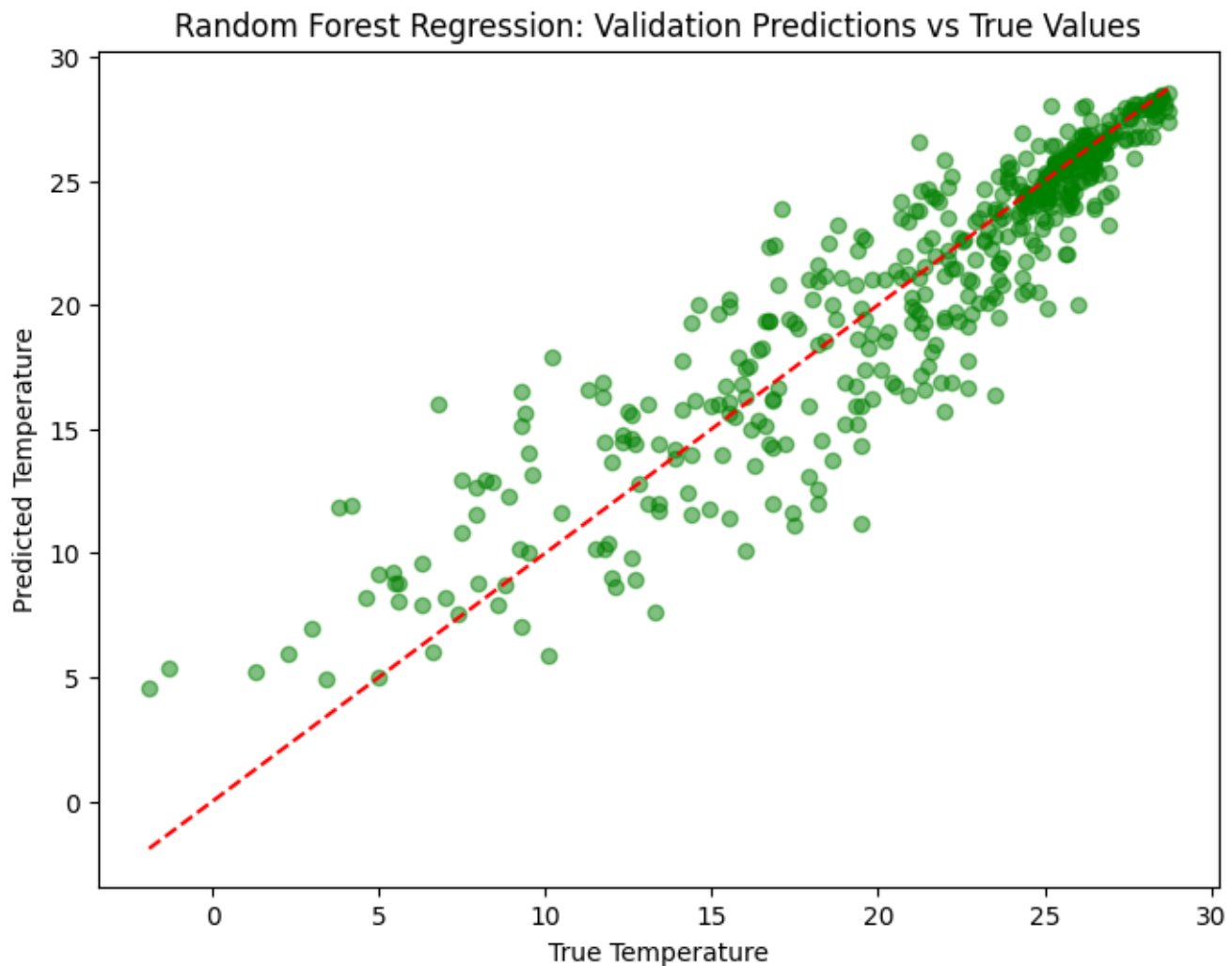


Metric	Accuracy
Validation Accuracy	0.9884
Test Accuracy	0.9851

The regression model code uses a Random Forest regressor, taking longitude and latitude as features to predict the temperature value at each valid grid point. The data is split into 70% training, 15% validation, and 15% test sets. The model learns on the training set, tunes and evaluates on the validation set, and finally assesses performance on the test set.

Model evaluation uses a scatter plot comparing predicted and true values on the validation set.

Metric	Mean Squared Error
Validation MSE	5.7138
Test MSE	5.3359



Overview

We took a bunch of gridded weather data, cleaned it up, and split it into three sets—training, validation, and test. We built two machine learning models: one to tell us which grid points have valid data, and another to predict the temperature at those valid points.

We also made two maps: a black-and-white one showing where the data is valid, and a colorful heatmap showing how temperature changes across the area. After training, we checked how well our models did using accuracy and mean squared error, and visualized the results with confusion matrices and scatter plots.